Unit: 6 Unit 6: Systems of Linear Equations and Inequalities **Name:** ____________
Topic: Applications of Systems (Word Problems) **Date:** __________

Homework (Habanero)

Q1. A pharmacy has x milligrams of Medication A and y milligrams of Medication B to create a new dosage. The total dosage needs to be 1200 milligrams, and the ratio of Medication A to Medication B needs to be $3:2$. What system of equations represents this situation?
(A) $x + y = 1200$, $x = 3y$
(B) $x + y = 1200$, $x = 3y/2$
(C) $2x + 3y = 1200$, $x = 3y$
(D) $2x + 3y = 1200$, $x = 3y/2$
(E) $x + y = 1200$, $2x = 3y$

Q2. A scientist is studying the growth of two types of cells. The population of Cell Type A grows at a rate of $2x$ cells per day, and the population of Cell Type B grows at a rate of $3y$ cells per day. If the total population is 1000 cells, and the ratio of Cell Type A to Cell Type B is $4:5$, what is the value of x?
(A) $x = 100$
(B) $x = 125$
(C) $x = 150$
(D) $x = 200$
(E) $x = 250$

Q3. A hospital is planning to administer a new vaccine. The vaccine requires x milliliters of Serum A and y milliliters of Serum B. The total amount of serum needed is 500 milliliters, and the ratio of Serum A to Serum B needs to be $2:3$. What is the value of y?
(A) $y = 150$
(B) $y = 200$
(C) $y = 250$
(D) $y = 300$
(E) $y = 350$

Q4. A doctor prescribes x grams of Medication A and y grams of Medication B to a patient. The total dosage needs to be 15 grams, and the ratio of Medication A to Medication B needs to be $1:2$. What system of equations represents this situation?
(A) $x + y = 15$, $x = y$
(B) $x + y = 15$, $x = 2y$
(C) $x + y = 15$, $2x = y$
(D) $x + 2y = 15$, $x = y$
(E) $2x + y = 15$, $x = 2y$

Q5. A researcher is studying the effects of x hours of exercise and y hours of sleep on the human body. The total amount of time spent on exercise and sleep is 10 hours, and the ratio of exercise to sleep needs to be $3:2$. What is the value of y?
(A) $y = 2$
(B) $y = 3$
(C) $y = 4$
(D) $y = 5$
(E) $y = 6$

Q6. A patient needs to take x milligrams of Medication A and y milligrams of Medication B to treat a disease. The total dosage needs to be 800 milligrams, and the ratio of Medication A to Medication B needs to be $5:4$. What system of equations represents this situation?
(A) $x + y = 800$, $x = 5y/4$
(B) $x + y = 800$, $x = 5y$
(C) $4x + 5y = 800$, $x = 5y$
(D) $4x + 5y = 800$, $x = 5y/4$
(E) $x + y = 800$, $4x = 5y$

Q7. A scientist is studying the growth of two types of bacteria. The population of Bacteria Type A grows at a rate of x cells per hour, and the population of Bacteria Type B grows at a rate of $2y$ cells per hour. If the total population is 500 cells, and the ratio of Bacteria Type A to Bacteria Type B is $3:5$, what is the value of x?
(A) $x = 50$
(B) $x = 75$
(C) $x = 100$
(D) $x = 125$
(E) $x = 150$

Q8. A hospital is planning to purchase x units of Medical Equipment A and y units of Medical Equipment B. The total cost needs to be

$$10,000$$

, and the ratio of Medical Equipment A to Medical Equipment B needs to be $2:3$. What system of equations represents this situation?
(A) $x + y = 10000$, $x = 2y/3$
(B) $x + y = 10000$, $x = 2y$
(C) $2x + 3y = 10000$, $x = 2y$
(D) $2x + 3y = 10000$, $x = 2y/3$
(E) $x + y = 10000$, $2x = 3y$

Open Ended Questions (FRQ)

Q9. A patient needs to take x milligrams of Medication A and y milligrams of Medication B to treat a disease. The total dosage needs to be 600 milligrams, and the ratio of Medication A to Medication B needs to be $2:3$. If the patient already took 100 milligrams of Medication A, how much more of each medication does the patient need to take?

Q10. A researcher is studying the effects of x hours of exercise and y hours of sleep on the human body. The total amount of time spent on exercise and sleep is 12 hours, and the ratio of exercise to sleep needs to be $4:5$. If the researcher wants to spend 2 more hours on exercise, how many hours should the researcher spend on sleep?

Chef's Tips

- Ensure students understand the context of each problem
- Remind students to check their units
- Encourage students to use graphs to visualize the solutions